BLUEPRINT — Week 11 — Team Bending (T4)

DRAFT for discussion, not an actual record of a live meeting. User, Money and Operator positions come from their submitted briefs. ***Technologist positions now come from Andrés's submitted brief*** *(`BRIEF_andres.pdf`; tests logged in `w11-fight/tests/`). Votes, declarations and dissent still need to be confirmed with every member.*

Team T4: Carlos Arakindji (USER) · Miguel Bravo (MONEY) · Nicolás Martínez (OPERATOR) · Andrés Álvarez Morphy (TECHNOLOGIST) · José Fernando Kuri (ADVERSARY)

FACTS WE AGREE ON

- Seeing losses is not the same as fixing them; a real-world response process is necessary. *Technologist evidence:* the city already knows it loses **42%** of its water to leaks (SACMEX/IPDP, 2022) and the Cutzamala is **92–94% full** (Oct-2026). The brief's 38% is stale. The gap is repair, payment and allocation, not sensing.
- Current tools can capture weights, issue digital receipts and detect water anomalies; verification and incentives are harder. *Technologist evidence (tested):*
  - Detecting "is there water" is commodity.
  - **Which** water is not: a generic audio model reads a toilet flush as water **93%** of the time (400 ESC-50 clips).
  - Vision models read **digital displays ~80%** but **dial gauges 10–32%** (MeasureBench 2025).
  - Self-reported photos are not proof: India found **~700,000 fake EPR certificates** (CPCB).
- Informal recyclers may lack documentation and bargaining power. Any solution must work without forcing an RFC, bank account or app download.
- International models exist: Colombia's recycler remuneration, Brazil's reverse-logistics credits, India's SWaCH, and water-monitoring platforms such as TaKaDu. Direct replication in Mexico faces institutional and financing barriers. *Technologist adds:* NextDrop (India, water-arrival alerts; RCT "null-to-modest", **63% of human reports wrong**) and Cataki (Brazil, catador app).

BETS WE CHOSE — PROPOSED, NOT VOTED

- Select pepenador transaction records rather than water-loss monitoring for the first pilot: one scrap yard can test it without municipal infrastructure contracts. **Proposed dissent: Money, User and Technologist prefer water visibility.** With the Technologist brief, 3 of 5 briefs lean to water, so this needs a real vote. If it ties, the Adversary breaks it by choosing the vacuum he failed to kill.
- Start with a voluntary brand-funded pilot, not presumed EPR revenue. The future legal payment channel is unverified. *Technologist confirms:* the Ley General de Economía Circular (DOF 19-ene-2026) still lacks its reglamento, national program and REP implementation agreement.
- Scrap-yard participation depends on incentives, but the recycler must retain control over their own records.

PRIMARY VACUUM

PEPENADOR RAILS — "Mi Boleta": a weight-and-price receipt from the existing yard scale to the recycler via WhatsApp or paper folio. **Backup: water loss-to-repair accountability**. The Technologist's version is the **delivery ledger at the tap**: SEGIAGUA publishes a schedule for 2,006 colonias; nobody measures whether water arrives. Already built: https://w11-llave.vercel.app.

THE CONDITIONS — THIS FUTURE MAY PROCEED, PROVIDED THAT…

1. No RFC, bank account, app installation or smartphone is required; paper receipt alternative.
2. Every receipt includes weight, material, unit price, total, date, yard and a dispute route; the worker gets a copy.
3. A named scrap-yard owner and a paying pilot sponsor commit before development; publish pilot cost and adoption metrics.
4. Validate weights against scale evidence and outbound invoices; flag possible fraud. Mass balance alone does not prove origin. *Technologist amendment (proposed):*
   - A photo of a **digital** scale display is evidence, not proof; analog dials are entered by hand.
   - Fraud is judged over **many transactions against an independent reference**, never on one.

    - A flag says "inconsistent", never the cause.
5. **SHADOW CLAUSE:** never score, rank, geotrack or penalize recyclers for refusing confirmation. Workers control personal history; brands see aggregated data. *Technologist amendment (proposed):*
    - **The measurement is not voted**: a worker's record counts for them without anyone confirming it (Moola's failure in the chapter).
    - **Every filter publishes whom it leaves out.** In Andrés's build, each of 8 verification filters tended to erase the worst-off party until this rule was added.
6. Test whether transparency improves net pay and dispute outcomes; stop if retaliation, exclusion or data misuse rises.

DECLARATIONS — PROPOSED FOR MEMBER APPROVAL

- **Carlos Arakindji (USER)** → test receipt usefulness and safety with real recyclers; condition 5.
- **Miguel Bravo (MONEY)** → model sponsor fees, yard incentives and net benefit per verified kilo; condition 3.
- **Nicolás Martínez (OPERATOR)** → pilot a 5-second ticket workflow at one yard; conditions 1–2.
- **Andrés Álvarez Morphy (TECHNOLOGIST)** → builds his surviving idea on the backup vacuum, **"Llave"**, already live at https://w11-llave.vercel.app:
    - It compares what SEGIAGUA publishes for each colonia with when water reaches the tap: a dumb pressure sensor, ML only to learn the valve circuit, no public number about any house. Sensors are SIMULADO until a field night.
    - Honors **condition 5** (no per-house output, the measurement is not voted, omissions published) and **condition 6** (kill criterion written before data).
    - For Mi Boleta he hands over the condition-4 technical checks (digital-display reading, many-transaction fraud test).
- **José Fernando Kuri (ADVERSARY)** → run fraud, retaliation and nonparticipation kill-tests; conditions 5–6.

DISSENT PRESERVED

- **Money:** water-loss repair has clearer avoided-cost metrics.
- **User:** public water telemetry could address more urgent household harms.
- **Adversary:** "Mi Boleta" dies without evidence of willing yards, fairer prices and protection from retaliation.
- **Technologist (Andrés):**
    - A phone photo of a receipt is the exact fraud surface India's EPR market collapsed on, so Mi Boleta's proof has to come from the scale and the yard's outbound invoices, not the photo.
    - The measurement he can stand behind this week is water delivery at the tap.
    - On the provocation he holds **"mal planteada"**: the city sees the *size* of its losses and doesn't pay to see where; people can't see whether the published water arrives. And his own seeing layer would reach first the colonias promised "continuo" (Álvaro Obregón 90%, Iztapalapa 3%) unless the tube's data, not demand, decides placement. Condition dated 19-nov.

Required next step: replace the proposed decisions with the actual team vote and record who agrees or dissents.